

WARWICK

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ES197 Projects - Electronics



# Project

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**Project Aim: Produce an educational tool for Year 1 students to learn analogue low pass filters.**

Tool specification:

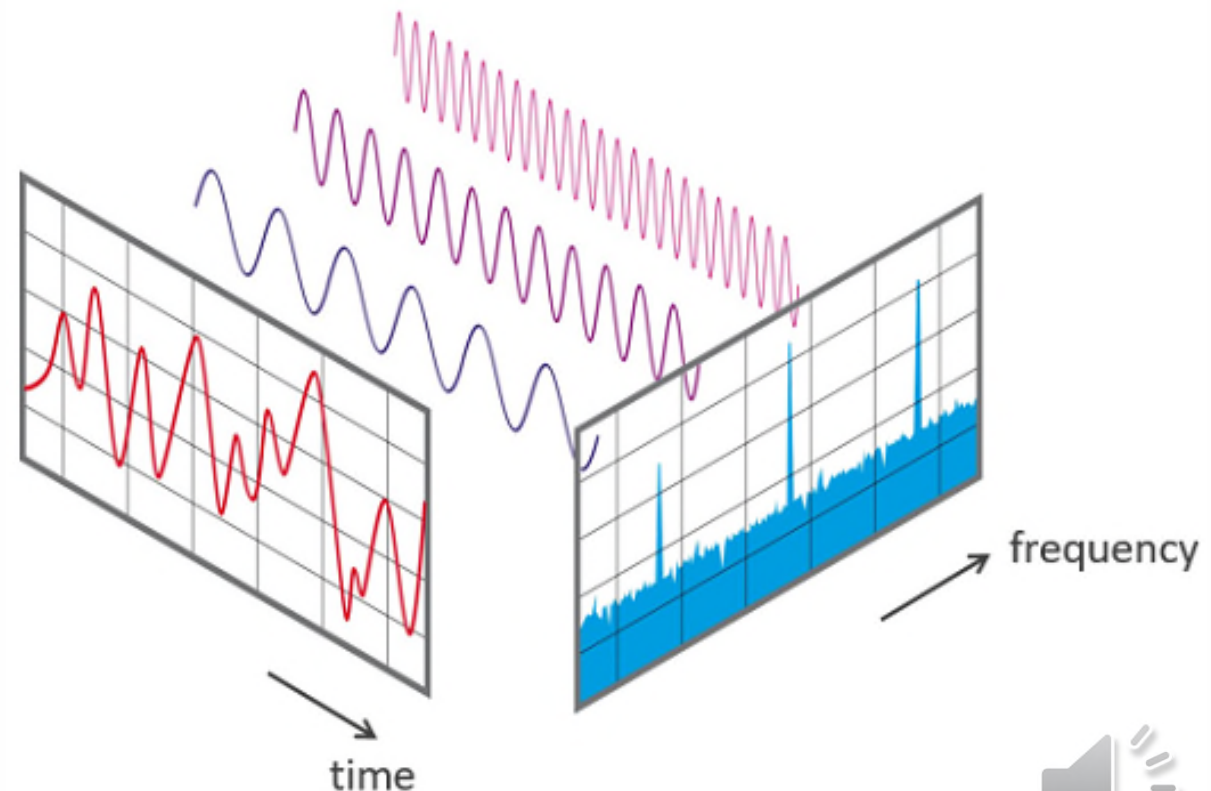
- Create a code to import a given sound file (with unwanted frequencies) to Matlab and identify its spectrum.
- Design and analyse a Butterworth filter circuit with the aid of Matlab to remove the unwanted frequencies. Compare the performance of the filter with a cascaded first order filter using Matlab/Simulink. All filter parameters (including circuit component parameters) and the filter theory on which the Matlab code(s) is (are) based, will be expected to be presented.
- Use a circuit simulator and compare its results to your Matlab/Simulink simulations.
- Build the filter circuits (using the breadboard and electronic components provided) and use the frequency response experimental results to confirm your simulated filter performance. You are provided with an arbitrary function generator, an oscilloscope and a power supply.
- Demonstrate experimentally your filters using the sound wave file given as input (play the sound using the PC) and use the PC headphone port and interfaces provided. You are provided with a portable loudspeaker and an oscilloscope.



# MATLAB - Fourier Transform

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- ▶ Left graph shows a signal in the time domain (red).
- ▶ Signal is made of three sine waves (purple).
- ▶ Applying the Fourier transform to the signal reveals the frequency and magnitude of purple waves.
- ▶ Right graph shows the signal spectrum in the frequency domain (blue).
- ▶ MATLAB - use `FFT()`



# Support information

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- (1) Mathematics for Engineers by Crofts and Davison, see chapter “An introduction to Fourier series”. Electronic copy available from the library.
- (2) Notes on Discrete Fourier Transformation (at the ES197 website) based on the book Signals and Systems by Meade and Dillon.
- (3) Practical Electrical Engineering, by Makarov et al, Section 9.3. Electronic copy available from the library.
- (4) ES191 moodle page.
- (5) Essentials of digital signal processing, B.P. Lathi and R. Green. Chapters 1 and 2 in particular. Available online from the library.
- (6) To realise filters (i.e. for filter circuits), search for resources on active filters.
- (7) Consult component datasheets ( e.g. <https://www.analog.com/media/en/technical-documentation/data-sheets/op177.pdf> )
- (8) The sound file can be found at the ES197 moodle page.



Attention

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**Please read carefully the ES197 Electronics Group Project Briefing provided at the ES197 moodle page including the Health and Safety information.**

